DAY 04: EC2 PURCHASING MODELS, LAMBDA, LIGHTSAIL & ELASTIC BEANSTALK

Cloud Computing & DevOps Master Curriculum

1. EC2 Purchasing Models

Model	Description & Best Use Case
On-Demand	Pay by second/hour with no long-term commitment. Best for unpredictable workloads & testing.
Spot Instances	Unused AWS compute capacity at up to 90% discount. Can be terminated with 2-min warning. Best for fault-tolerant batch processing.
Reserved Instances (RI)	Commitment to 1 or 3 year term for up to 72% discount. Best for steady-state applications.
Savings Plans	Flexible pricing model commitment ($/hr spend) offering up to 72% savings across EC2, Fargate, and Lambda.
Dedicated Hosts	Physical server dedicated to your use. Best for strict compliance or existing per-socket software licenses.

2. AWS Lambda (Serverless Compute)

AWS Lambda allows you to run code without provisioning or managing servers. You pay only for the compute time consumed (per millisecond).

Triggered automatically by events from AWS services like S3 uploads, DynamoDB updates, HTTP API Gateway requests, or EventBridge schedules.

3. Amazon Lightsail vs AWS Elastic Beanstalk

Amazon Lightsail: Simplified virtual private server (VPS) platform designed for small businesses, blogs, and simple websites. Bundles compute, storage, and networking into predictable monthly pricing.

AWS Elastic Beanstalk: Platform as a Service (PaaS) for deploying and scaling web applications developed with Java, .NET, PHP, Node.js, Python, Ruby, Go, and Docker. You upload code; Beanstalk automatically handles provisioning, load balancing, auto-scaling, and health monitoring.

KEY TAKEAWAYS & SUMMARY

✔ Spot Instances dramatically reduce compute cost for interruptible batch workloads.

✔ AWS Lambda enables event-driven serverless architecture with zero server management.

✔ Elastic Beanstalk provides automated PaaS deployment while giving full control over underlying AWS resources.